



GOOGLE APPS PERFORMANCE

Consultant Dashboard

Imagine you are a Consultant preparing a market research for a mobile app company.
One of your daily tasks is to **meet the Product Manager and evaluate their App idea.**

My **Consultant Dashboard** gives you a practical way
to walk app creators through the market data behind their ideas,
helping you ground the conversation in clear, **data-driven insights.**



git



GitHub

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Introduction

Imagine you are a **Market Research Consultant** hired by a mobile app company. One of your tasks is to assist the **Product Manager** in evaluating their app ideas.

App creators need **fast, objective insight** into how their concept fits the current mobile-app landscape.

This Consultant Dashboard provides exactly that.

It helps the Consultant to:

- present the **most competitive** app categories;
- showcase the specific category targeted by the creators;
- **compare their goals against market medians.**



With these inputs, the Product Manager can ground their decisions in data rather than assumptions and quickly understand where their app stands relative to the market.

Data source

Imagine you are a **Market Research Consultant** hired by a mobile app company. One of your tasks is to assist the **Product Manager** in evaluating their app ideas.

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Google Play Store Apps dataset via Kaggle

Authored by L. Gupta (2019) [[link](#)]

```
{ googleplaystore.csv }  
{ googleplaystore_user_reviews.csv }
```

Approach

O1: Dataset preprocessing

O1.1 Loading both datasets

O1.2 Quality check and transforming variables

O1.3 Merging both datasets into one dataframe

O1.4 Missing data review and saving results

Python {pandas, matplotlib, seaborn}

Git & GitHub for version control

Examples of data preprocessing solutions

```
print("---- 01.2 Performing data quality checks and variable transformations...\n")
print()
print("Main dataset variable anomalies:")
print(f"• Categories: {len(df_dat.Category.unique())} unique values")
print(f"• Found anomaly in Category column: '1.9'")
print(f"• For abnormal row all values were shifted left between columns.")
df_dat[df_dat.columns[1:]] = df_dat[df_dat.columns[1:]].astype('object')
df_dat.loc[df_dat['Category'] == '1.9', df_dat.columns[1:]] = (
    df_dat.loc[df_dat['Category'] == '1.9', df_dat.columns[1:]].shift(1, axis=1))
print("✓ Anomaly corrected\n")
```

Quality checks: I inspected dataset columns and their content to catch potential anomalies in data types. Here I found data entry and corrected them to avoid losing otherwise correct entries.

```
df_dat['Last Updated'] = pd.to_datetime(df_dat['Last Updated'],
                                       format = '%B %d, %Y', errors = 'coerce')
nan_pct = df_dat['Last Updated'].isnull().sum()*100/len(df_dat['Last Updated'])
print(f"• Last Updated → datetime: {nan_pct:.1f}% values converted to NaT")
```

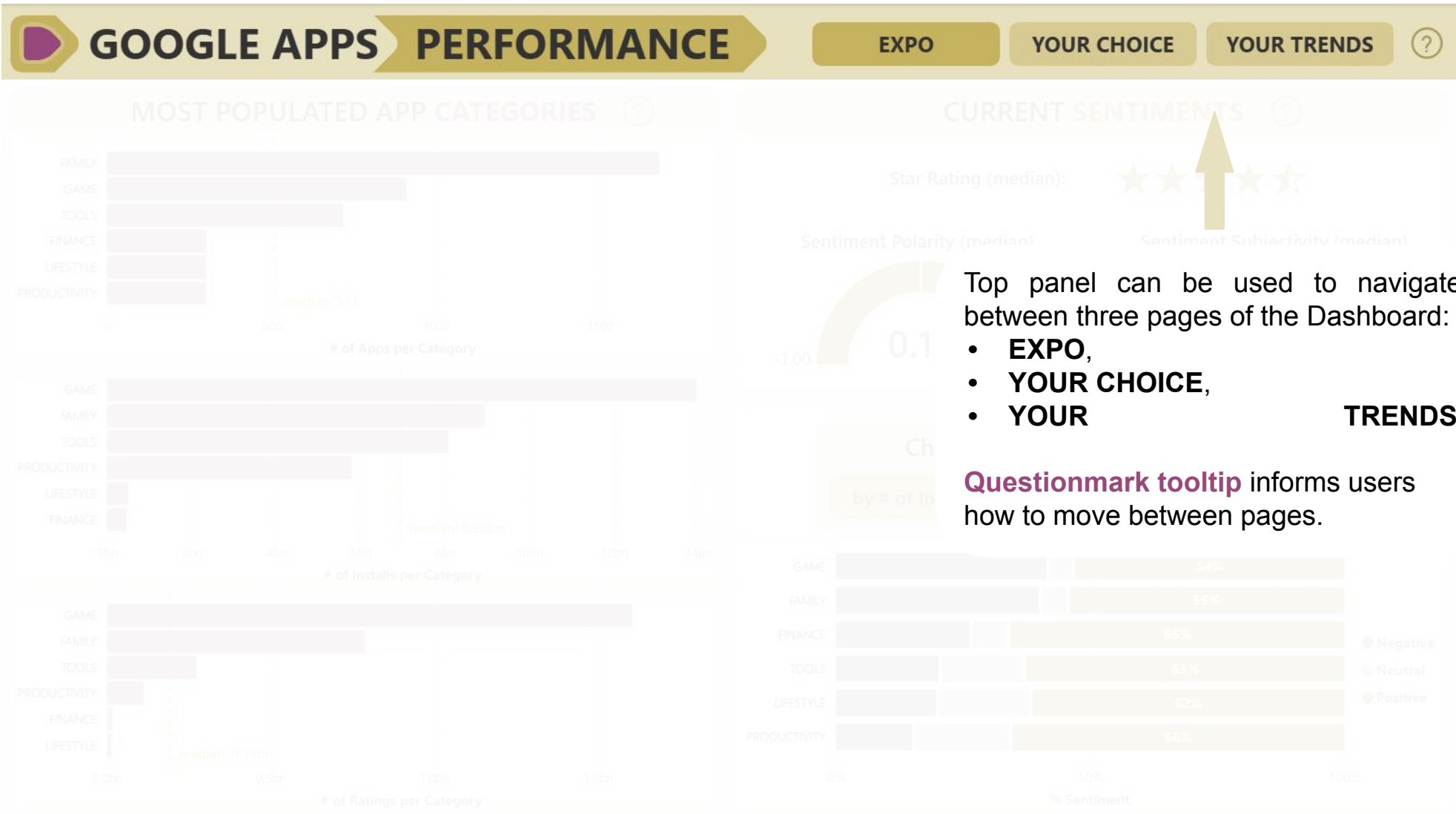
Transforming variables: I transformed the 'Last Updated' column into a correct date-time format and displayed the percentage of entries that could potentially be lost due to conversion.

```
print("---- 01.4 Missing data analysis:")

nan_count = df.isna().sum()
all_count = df.iloc[:,0].count()
prc = (nan_count * 100)/all_count
for col, pct in prc.items():
    if pct > 0:
        print(f"• {col}: {pct:.0f}% missing")
```

Missing data review: I calculated and evaluated missing data, column-wise. Knowing the structure and characteristics of both datasets, I diagnosed the cause of data entry gaps.

Creating the Dashboard

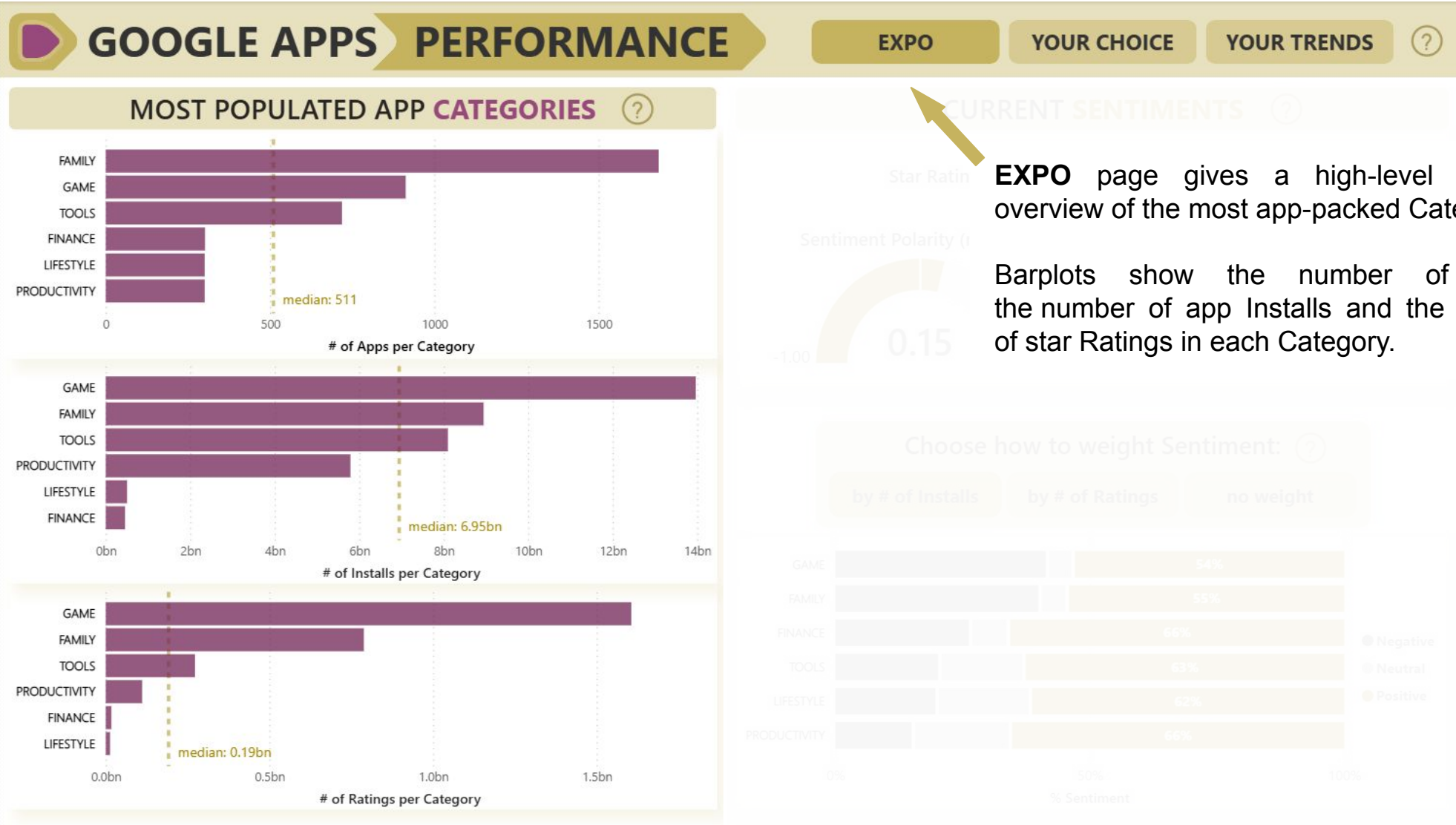


Top panel can be used to navigate between three pages of the Dashboard:

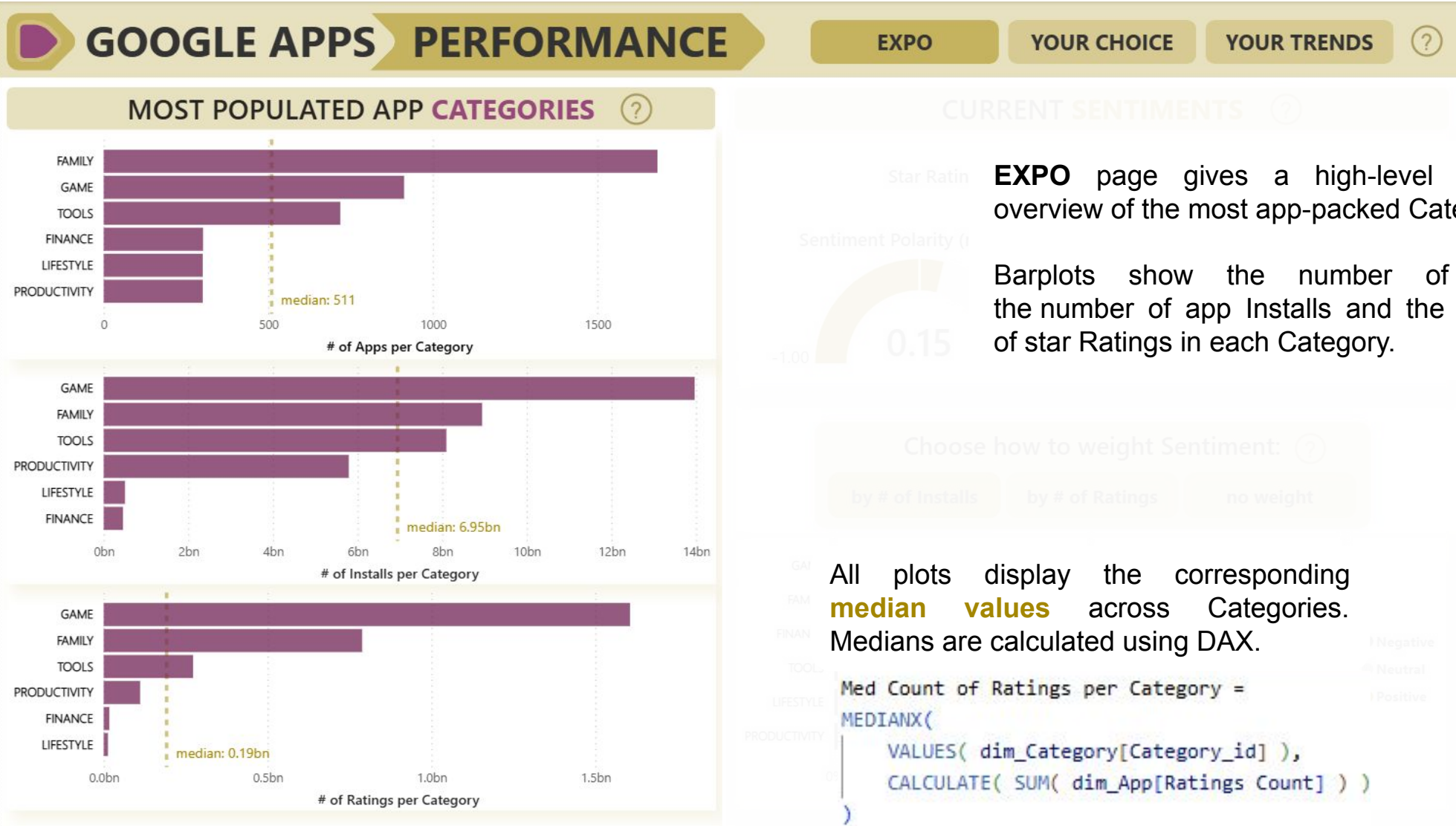
- **EXPO,**
- **YOUR CHOICE,**
- **YOUR TRENDS.**

Questionmark tooltip informs users how to move between pages.

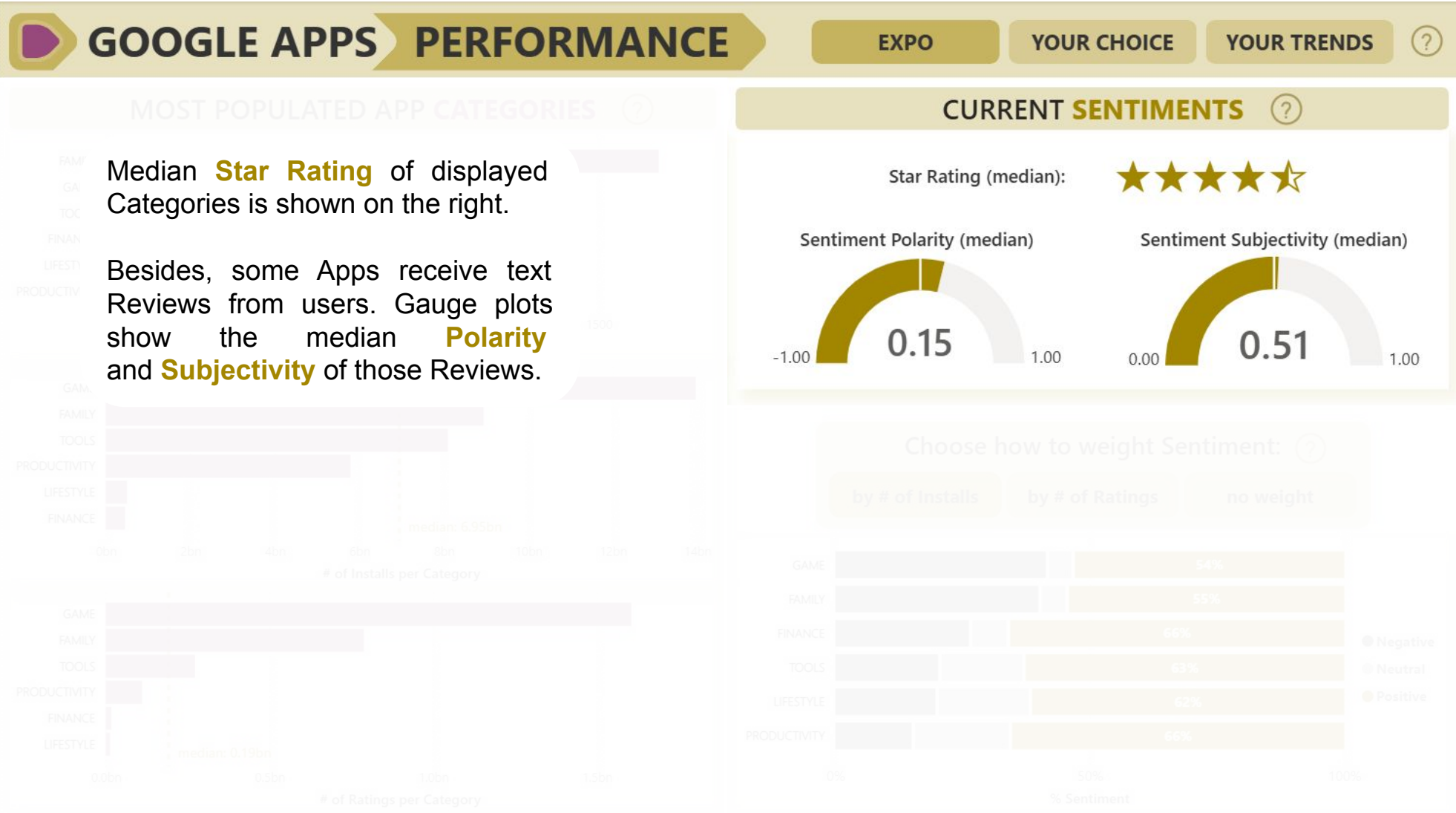
Creating the Dashboard



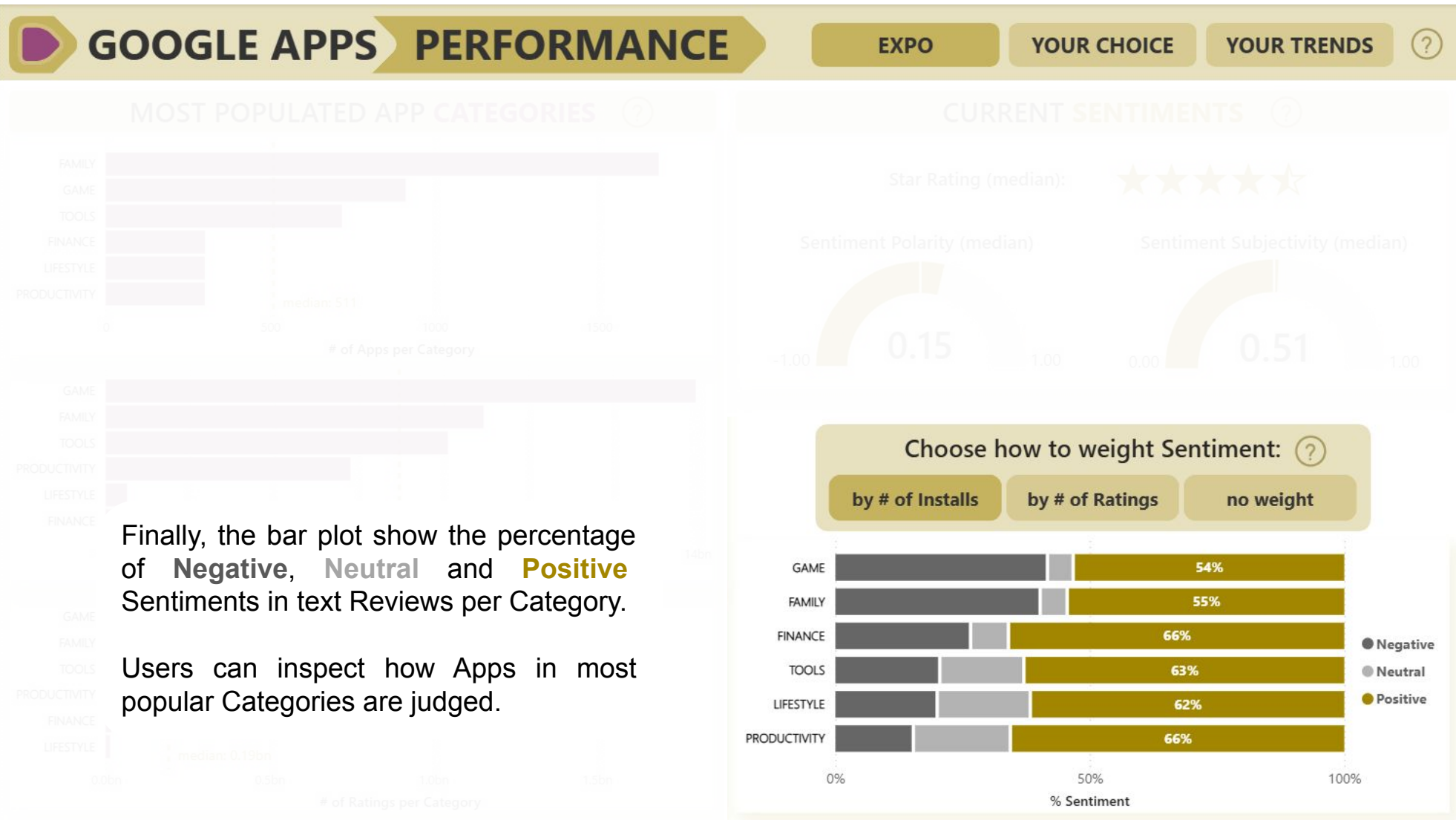
Creating the Dashboard



Creating the Dashboard



Creating the Dashboard



Finally, the bar plot show the percentage of **Negative**, **Neutral** and **Positive** Sentiments in text Reviews per Category.

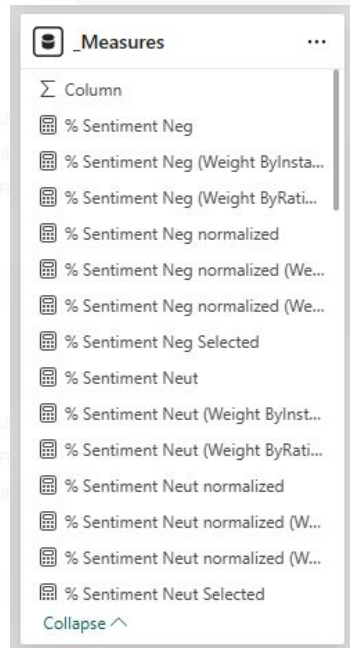
Users can inspect how Apps in most popular Categories are judged.

Creating the Dashboard

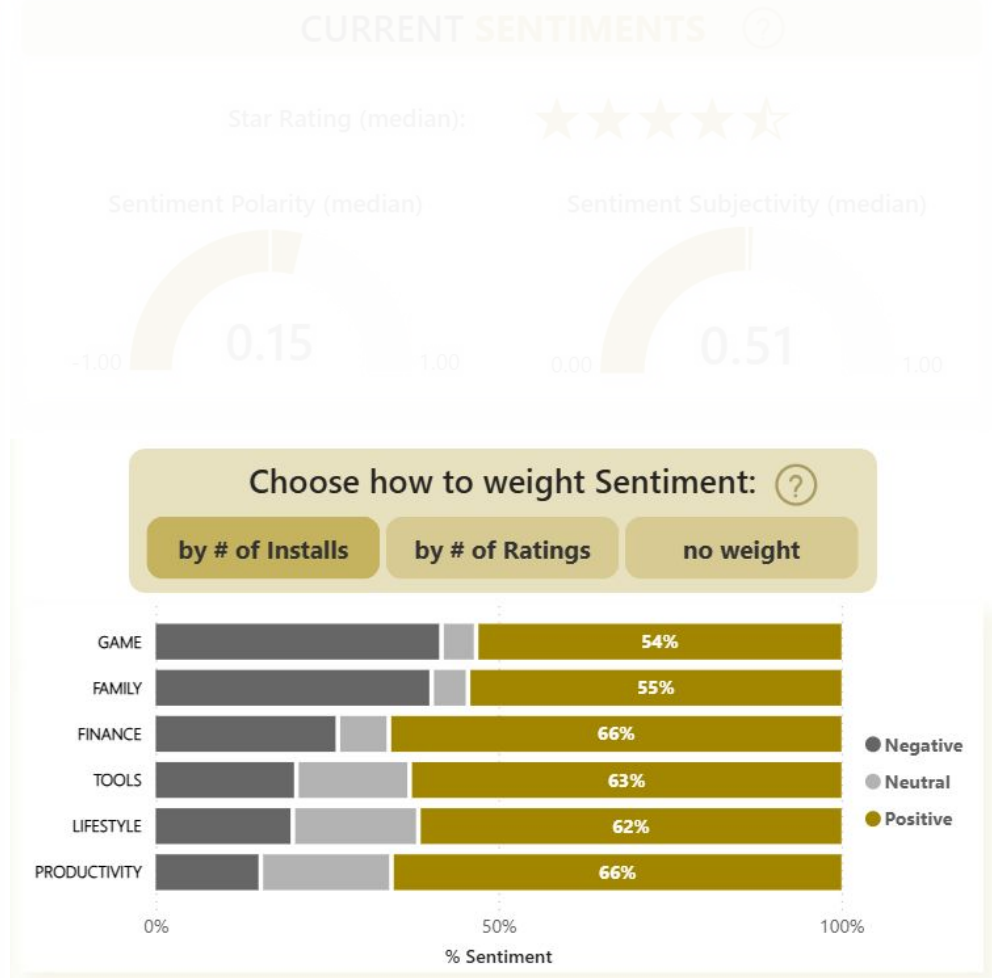
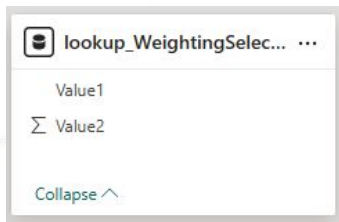


Those percentages are normalized and can be:

- weighted by the number of Installs*;
- weighted by the number of Star Ratings;
- unweighted.



Custom **DAX measures** and **tables** were created to maintain this solution.



* Apps with high Installs count heavily influence the average.

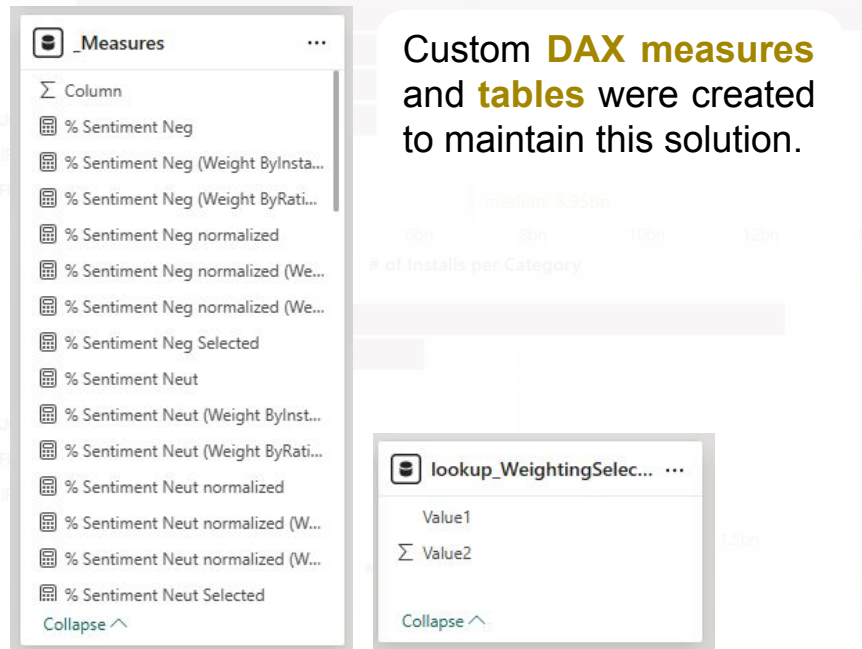
Creating the Dashboard



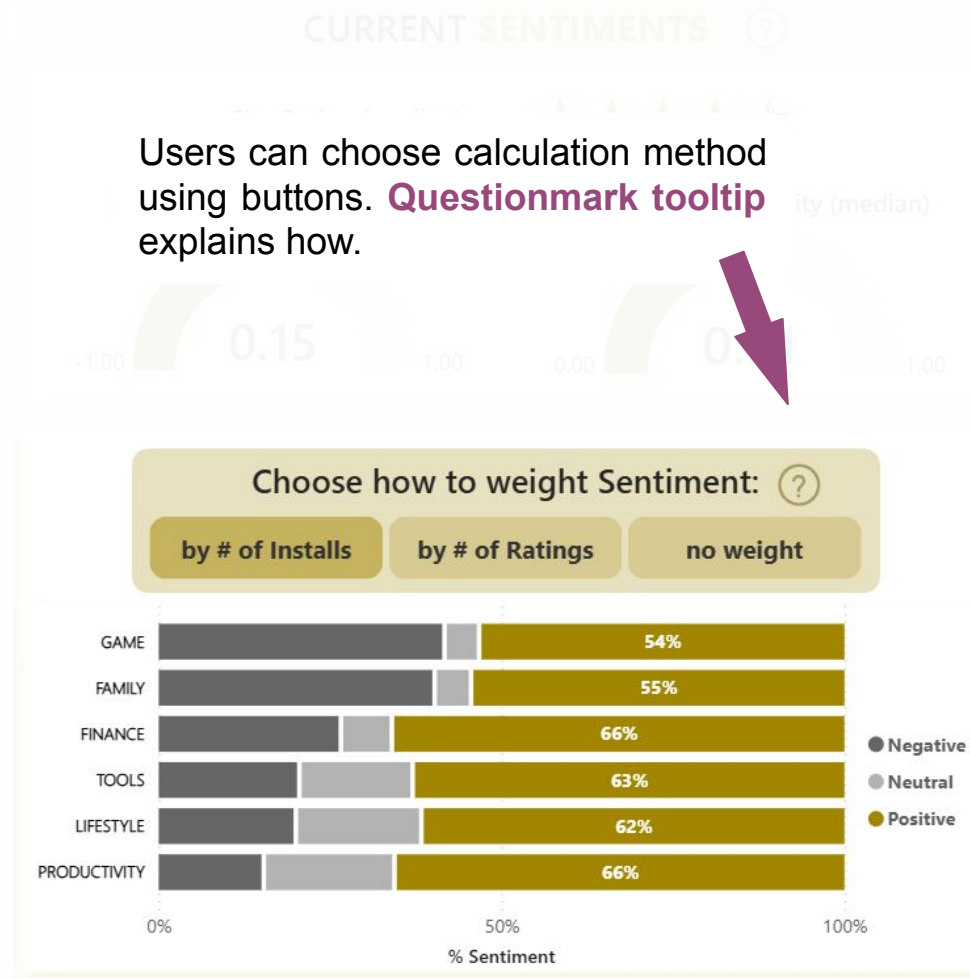
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- weighted by the number of Installs*;
- weighted by the number of Star Ratings;
- unweighted.

Users can choose calculation method using buttons. **Questionmark tooltip** explains how.

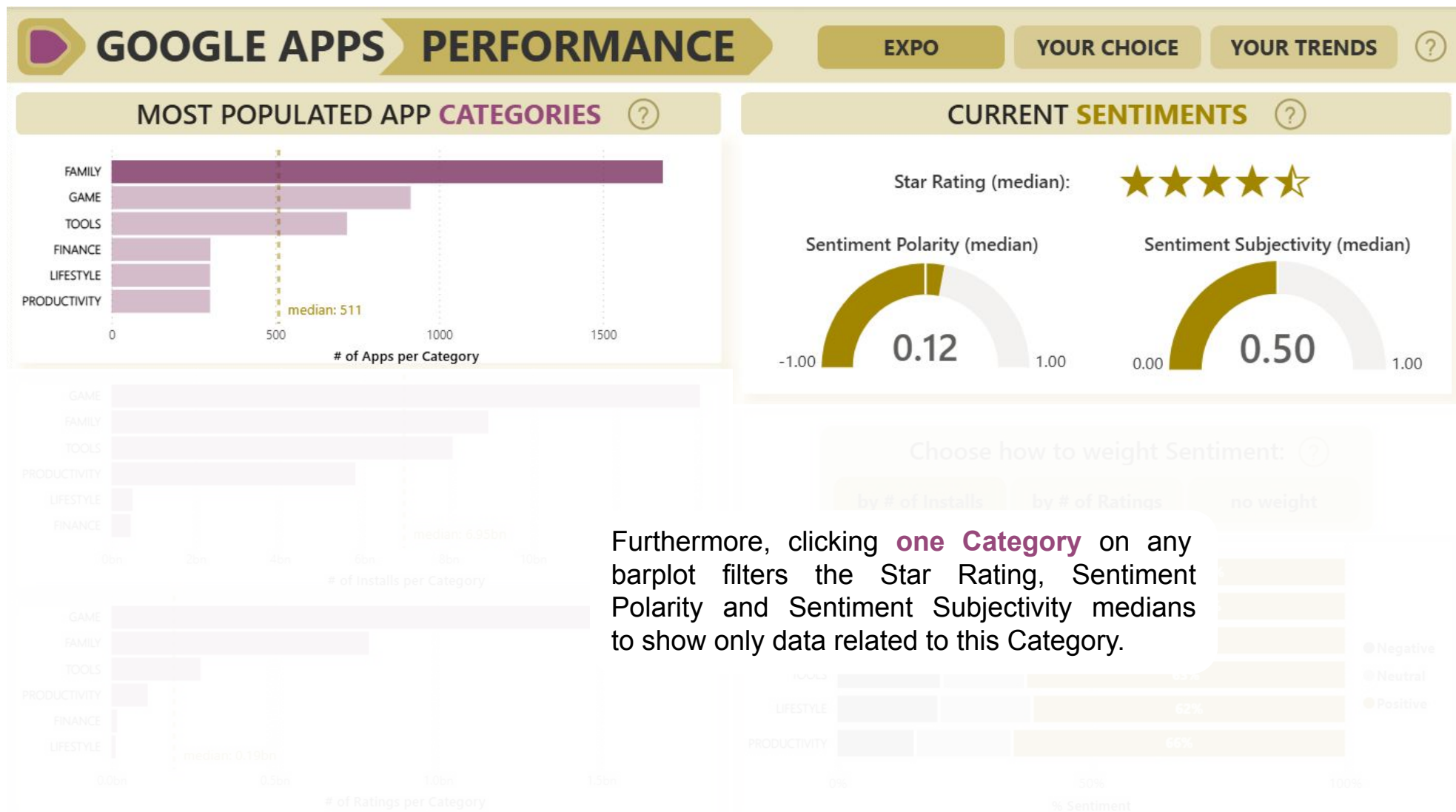


Custom **DAX measures** and **tables** were created to maintain this solution.

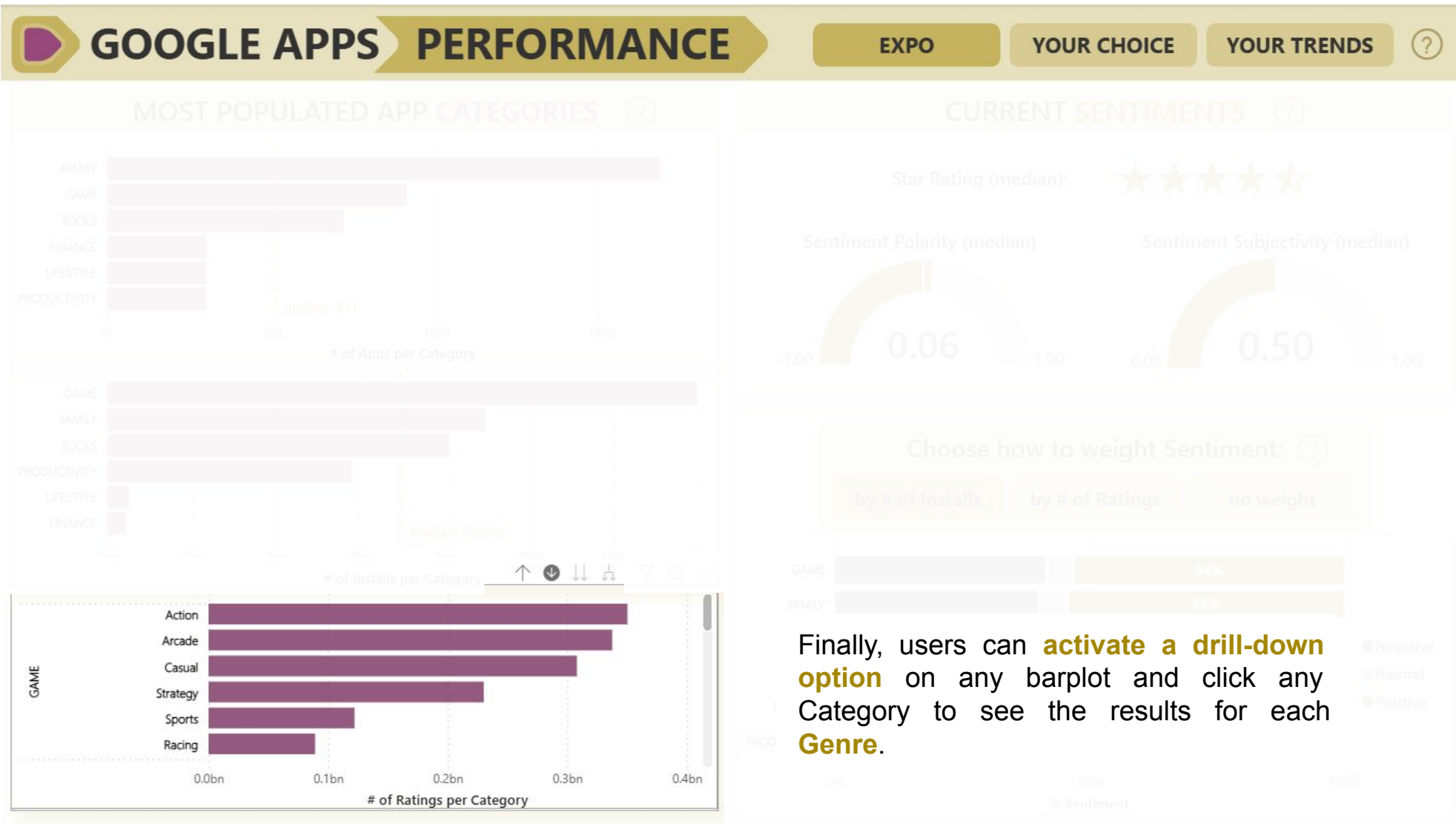


* Apps with high Installs count heavily influence the average.

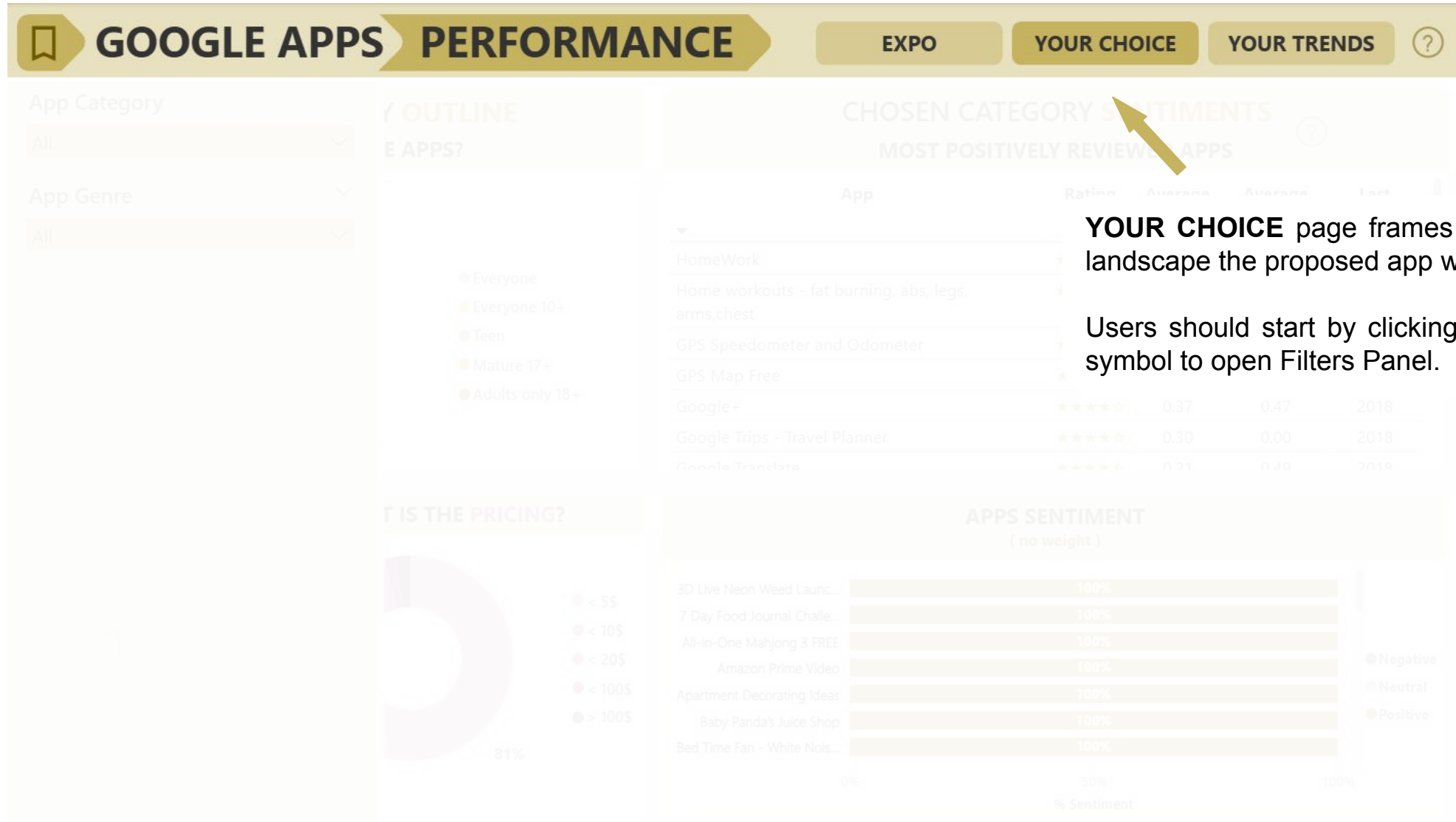
Creating the Dashboard



Creating the Dashboard



Creating the Dashboard



YOUR CHOICE page frames the competitive landscape the proposed app would enter.

Users should start by clicking the **Bookmark** symbol to open Filters Panel.

Creating the Dashboard

The screenshot shows the Google Apps Performance dashboard. At the top, there's a navigation bar with a bookmark icon, the title "GOOGLE APPS PERFORMANCE", and three tabs: "EXPO", "YOUR CHOICE", and "YOUR TRENDS". Below the navigation bar, on the left, is the "Filters Panel" with two dropdown menus: "App Category" (set to "All") and "App Genre" (set to "All"). To the right of the filters panel, the main content area displays various charts and tables. A table titled "VIEWED APPS" is visible, showing columns for "Rating", "Average Polarity", "Average Subjectivity", and "Last Updated". The "Filters Panel" is currently open, showing a list of app categories and genres. The "App Category" dropdown is set to "EDUCATION". The "App Genre" dropdown is set to "Education;Creativity". The "Filters Panel" can be closed by clicking the bookmark icon in the top navigation bar.

In Filters Panel users can choose **App Category** that is of interest to the Product Manager.

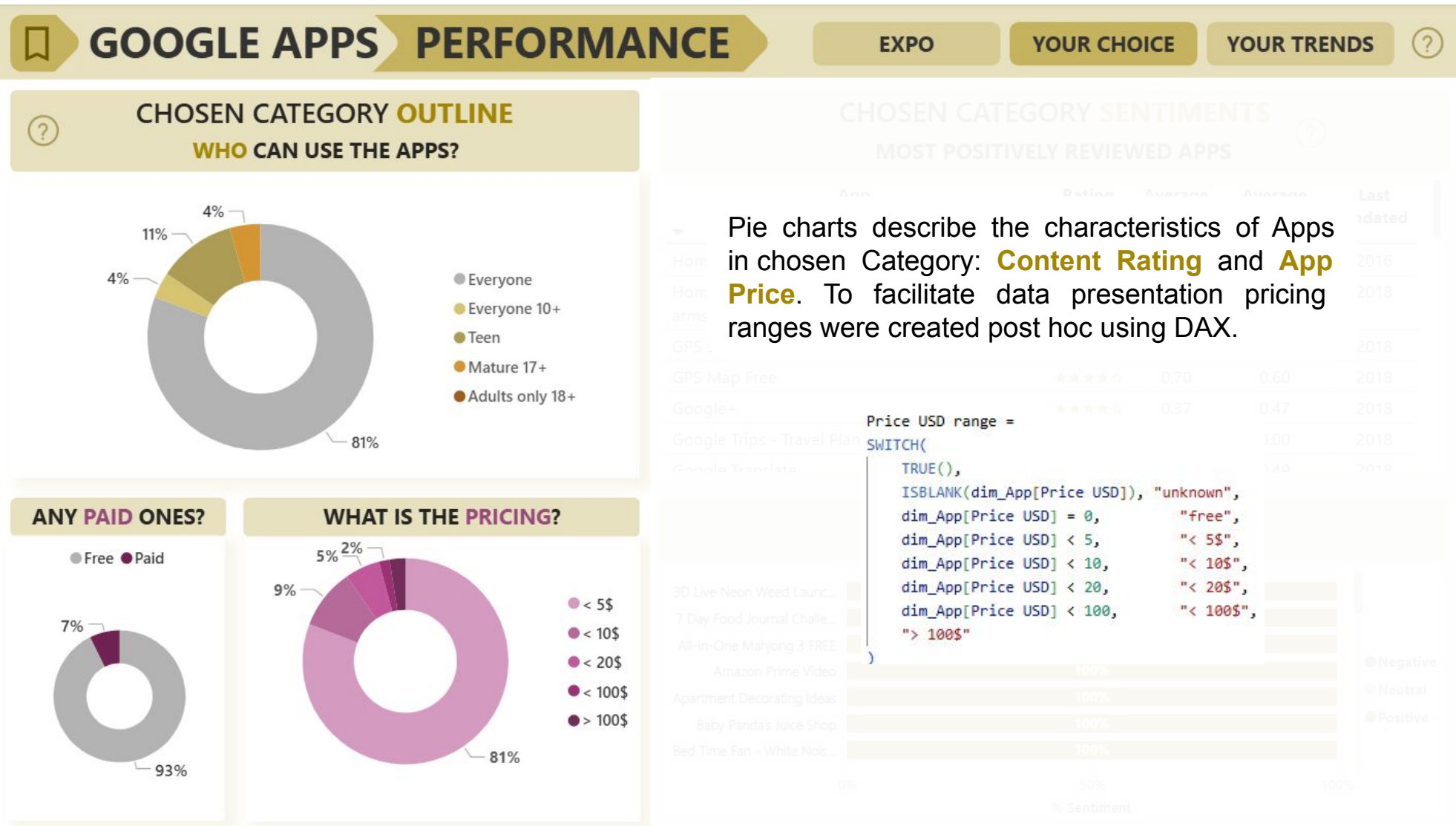
Some Categories have several **Genres** (category sub-types) that can be selected to narrow down the focus of the analysis.

Example:

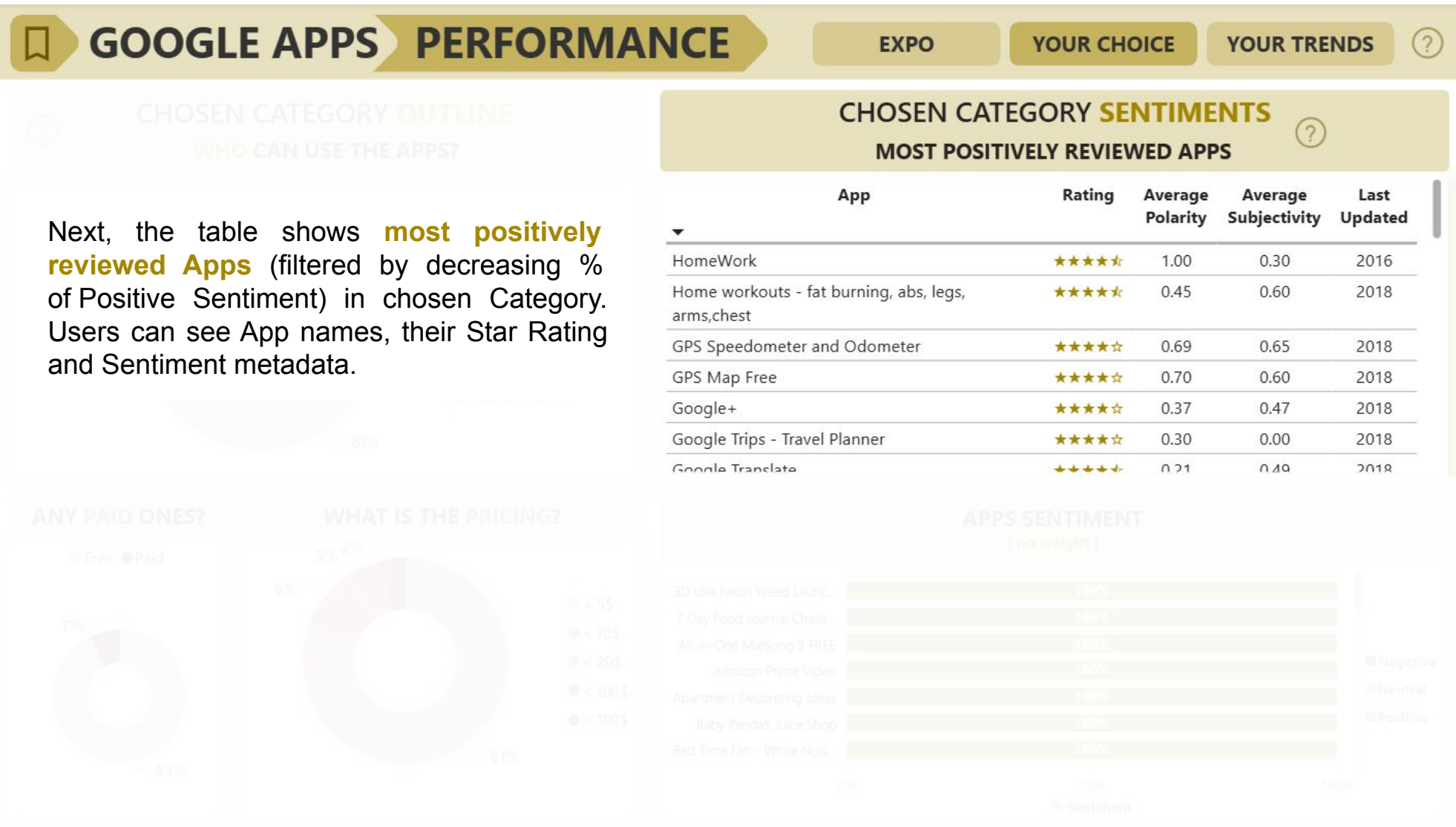
- App Category: EDUCATION
- App Genre: Education;Creativity

Filter Panel can be closed by clicking the **Bookmark** symbol again.

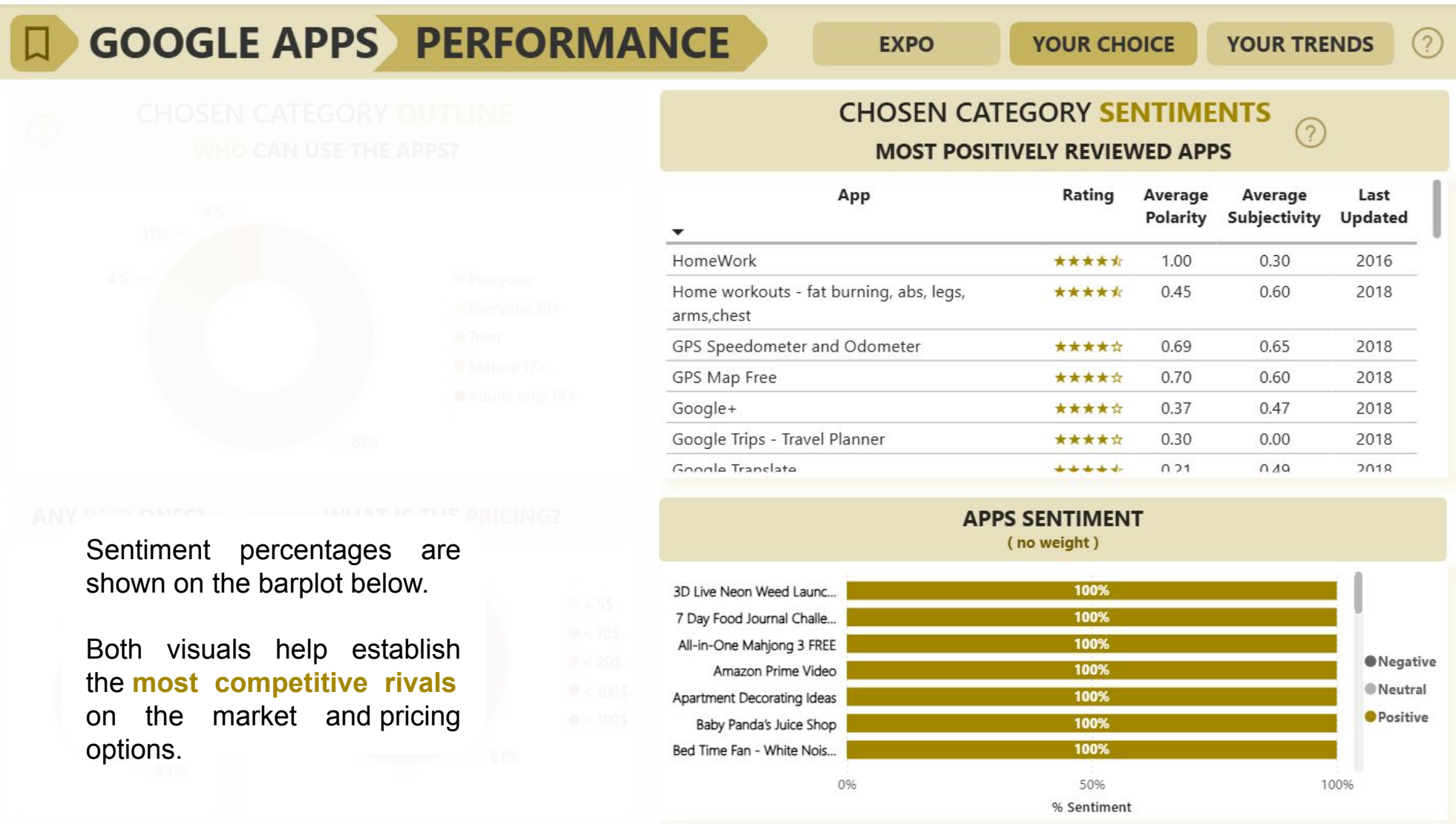
Creating the Dashboard



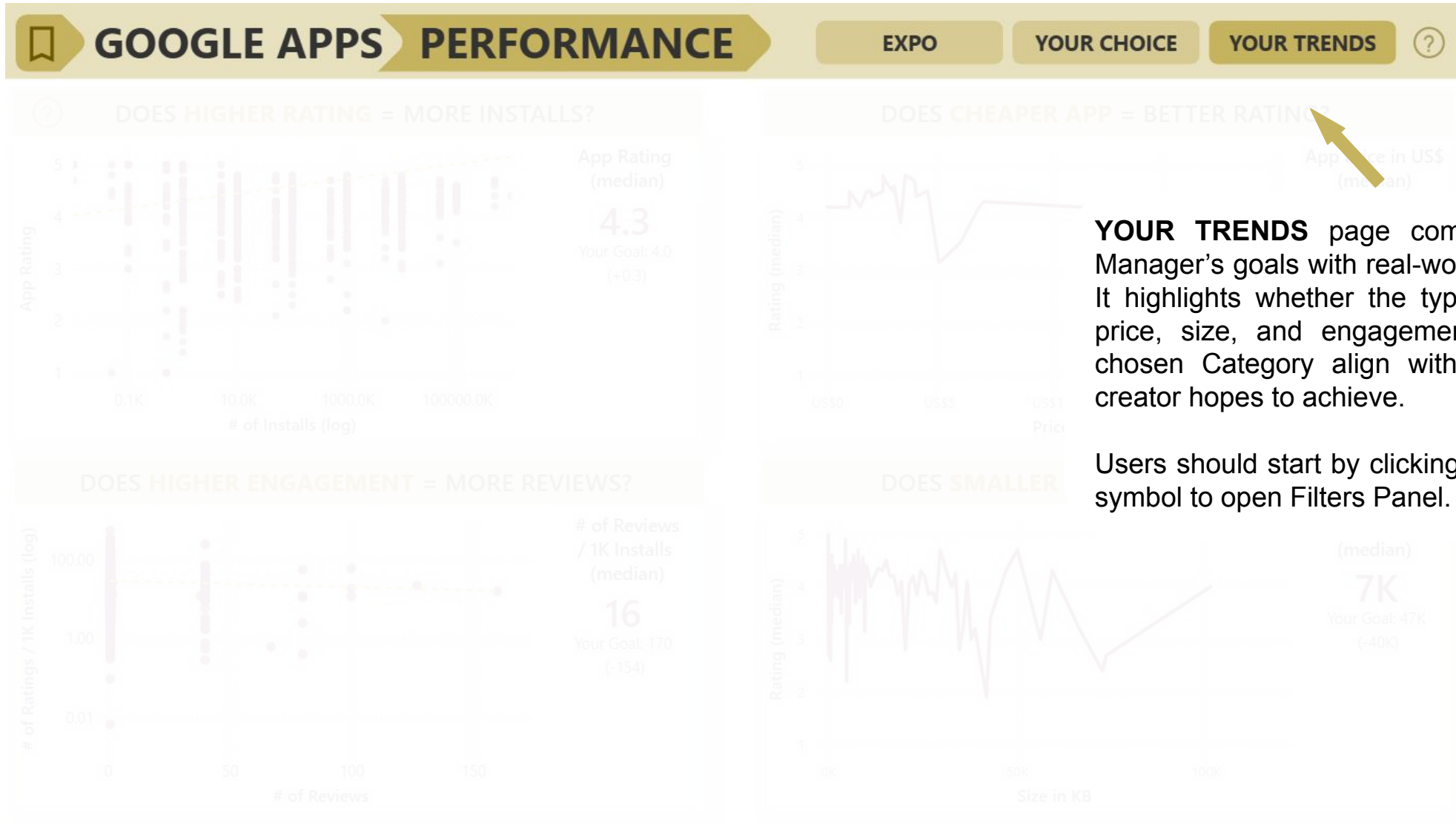
Creating the Dashboard



Creating the Dashboard



Creating the Dashboard



YOUR TRENDS page compares Product Manager's goals with real-world benchmarks. It highlights whether the typical App rating, price, size, and engagement levels within chosen Category align with what the app creator hopes to achieve.

Users should start by clicking the **Bookmark** symbol to open Filters Panel.

Creating the Dashboard



In Filters Panel users can choose **App Category**, **Genre** and target group (**Content Rating**).

Furthermore, several numerical KPIs (**Goals**) can be set to inform comparisons. For example, Product Managers may already have an idea of the **target App Rating** the product should achieve.

Creating the Dashboard



GOOGLE APPS PERFORMANCE

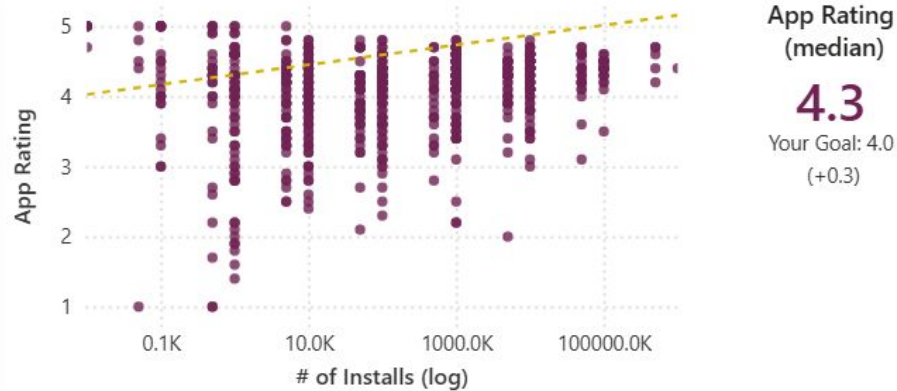
EXPO

YOUR CHOICE

YOUR TRENDS



DOES **HIGHER RATING** = MORE INSTALLS?



DOES **CHEAPER APP** = BETTER RATING?

The page includes four plots showing the relationships between benchmark performance measures.

For example, we can observe **moderate trend (dashed line)** of the number of App Installs increasing together with App Rating.

Price in US\$ (median)

0.0

Your Goal: 10.0 (-10.0)

DOES **HIGHER ENGAGEMENT** = MORE REVIEWS?



DOES **SMALLER APP** = BETTER RATING?

A number to the right is **median App Rating** for all Apps within previously chosen group (Category, Genre, Content Rating).

Importantly, this value is then compared to **goal App Rating**, to show how far does the target result stand from the real-world median.

Size in KB (median)

7K

Your Goal: 47K (-40K)

Creating the Dashboard



GOOGLE APPS PERFORMANCE

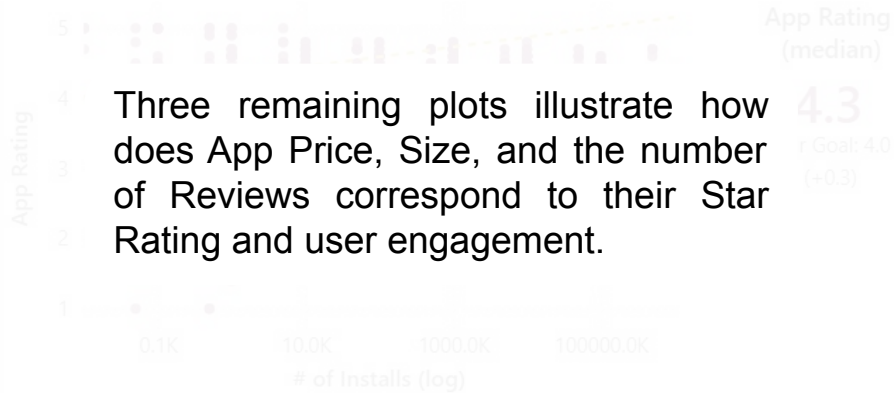
EXPO

YOUR CHOICE

YOUR TRENDS



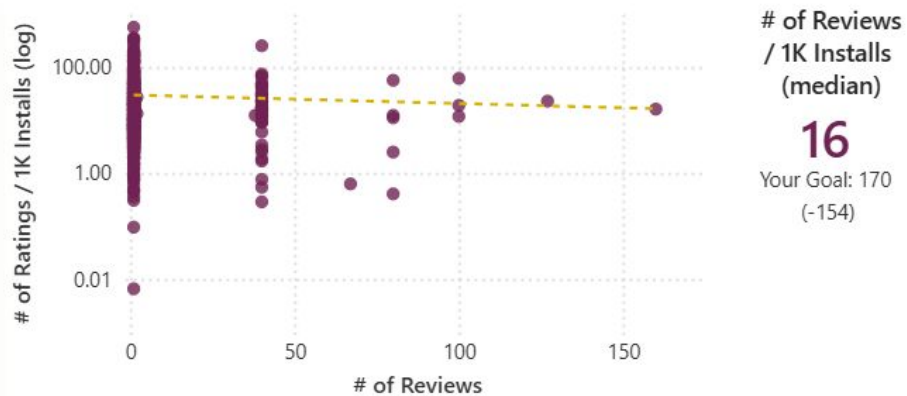
DOES HIGHER RATING = MORE INSTALLS?



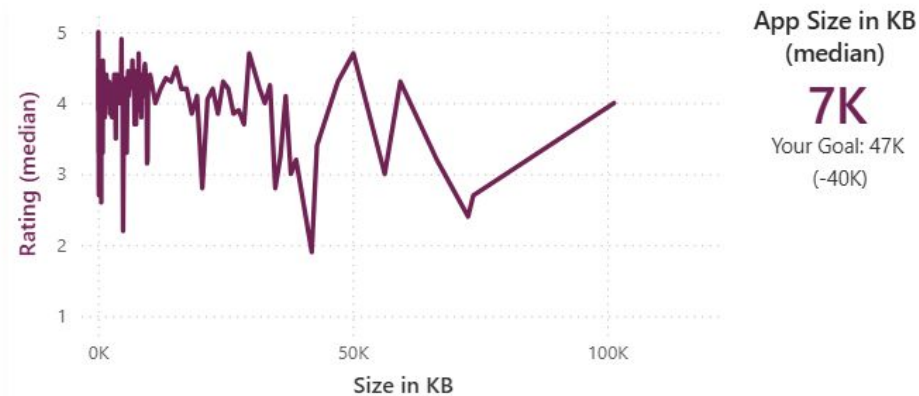
DOES **CHEAPER APP** = BETTER RATING?



DOES **HIGHER ENGAGEMENT** = MORE REVIEWS?



DOES **SMALLER APP** = BETTER RATING?



Creating the Dashboard



GOOGLE APPS PERFORMANCE

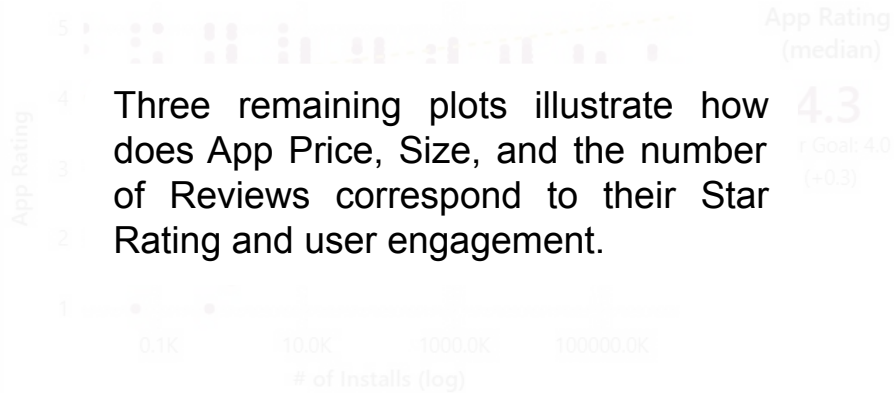
EXPO

YOUR CHOICE

YOUR TRENDS



DOES HIGHER RATING = MORE INSTALLS?



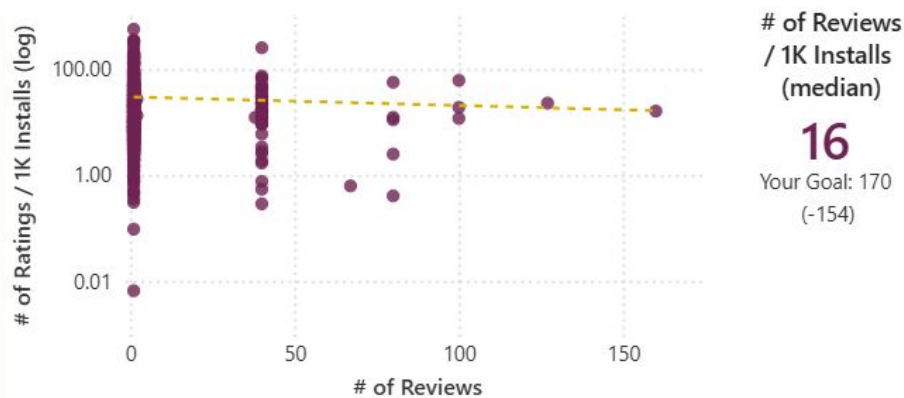
Three remaining plots illustrate how does App Price, Size, and the number of Reviews correspond to their Star Rating and user engagement.

Custom **DAX measures** and **tables** were created to maintain this solution.

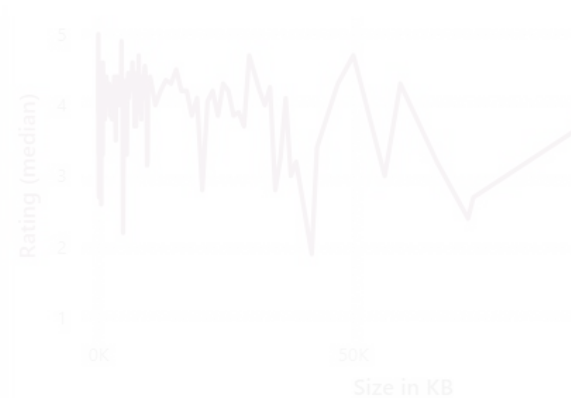
BETTER RATING?



DOES HIGHER ENGAGEMENT = MORE REVIEWS?



DOES SMALLER APP = BETTER RATING?



goal_AppRating

goal_AppRating

goal_AppRating Value

Collapse ^

goal_Ratings/1KInstalls

goal_Ratings/1KInstalls

goal_Ratings/1KInstalls Value

Collapse ^

goal_AppPrice

goal_AppPrice

goal_AppPrice Value

Collapse ^

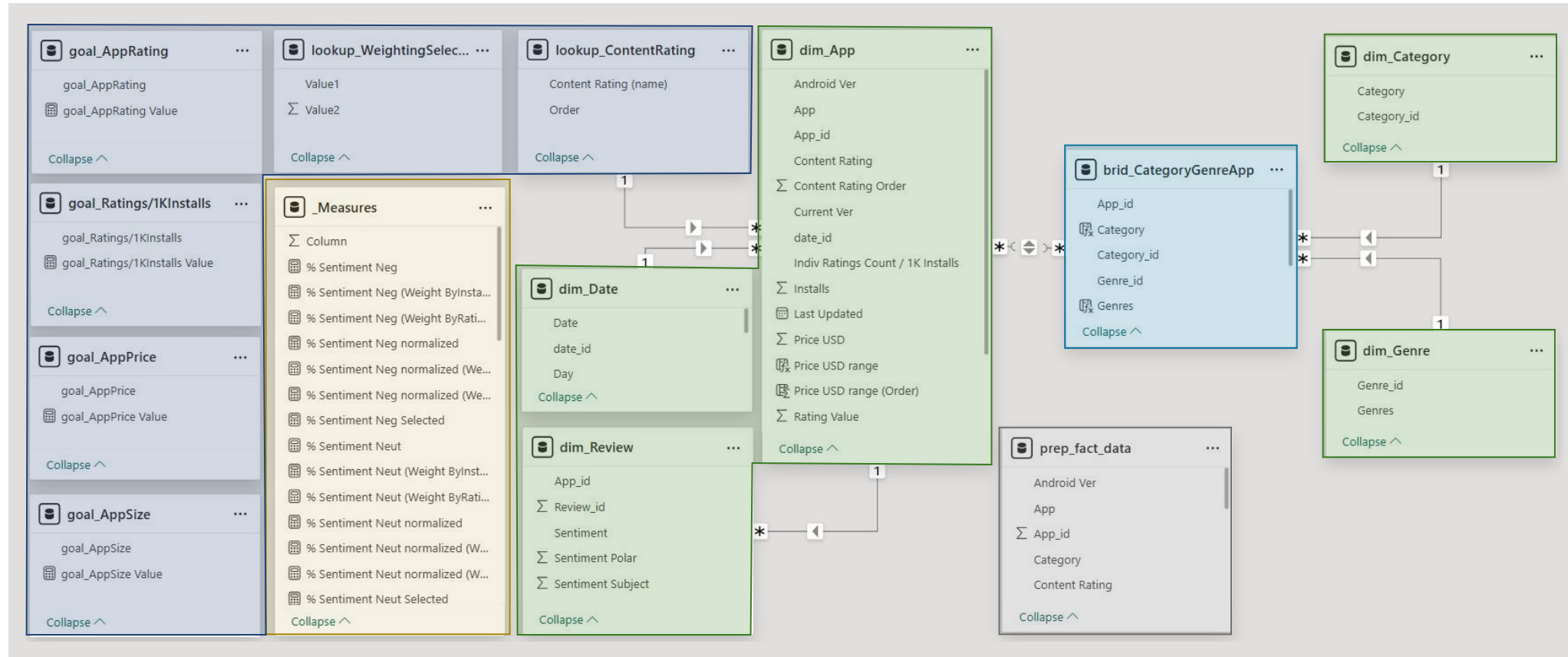
goal_AppSize

goal_AppSize

goal_AppSize Value

Collapse ^

Final Dashboard Model



Legend:

- DAX *_Measures* ensuring correct calculations across filters.
- Additional (*lookup_* and *goal_*) tables added to maintaining visuals and store input values.
- Dimentional (*dim_*) tables storing the source data and IDs.
- Bridging (*brid_*) table maintaining proper hierarchy of App–Genre–Category.
- Original source data table (*fact_*).

Final results



prep_fact_data.csv

*the main dataset merged from both Kaggle data sources
{googleplaystore.csv, googleplaystore_user_reviews.csv}
and preprocessed using Python codes.*



preprocessing_code.py

*the Python code that can be re-used in future
in case of a database update.*



Consultant_Dashboard.pbix

*the PowerBI dashboard based on the preprocessed dataset
{prep_fact_data.csv}*

Limitations of data coverage



prep_fact_data.csv

the main dataset merged from both Kaggle data sources {googleplaystore.csv, googleplaystore_user_reviews.csv} and preprocessed using Python codes.



preprocessing_code.py

the Python code that can be re-used in future in case of a database update.



Consultant_Dashboard.pbix

the PowerBI dashboard based on the preprocessed dataset {prep_fact_data.csv}

- The **Last Updated** field does not extend beyond 2018, indicating that the source dataset is no longer current and may not fully reflect today's app-store landscape.
- Several **Genres** contain only a small number of Apps, which limits the reliability of some category-level comparisons or trend-based interpretation.

Important note:

This dashboard was created as a portfolio project to demonstrate analytical and data-modelling skills, as well as the Power BI proficiency. While the dataset has clear limitations, it serves adequately for those purposes.

Key skills

Data modelling

- Designing a **star-schema model** (fact table + dimension tables).
- Creating one-to-many and one-to-one **relationships**.
- Resolving ambiguity with **bridging tables** (App–Genre–Category).
- Creating **calculated columns** (for categorical ranges and ordering).
- Building **aggregation** measures (counts, medians, ratios).

Data visualization

- Designing charts with **drill-down hierarchies** (Category → Genre).
- Configuring scatterplots with **logarithmic scaling** for a clean data presentation.
- Creating **user inputs** and **KPI cards** with goal-vs-median logic.
- Creating multi-page **thematic layouts** with **page navigation** and **filter panels**.

Storytelling

- Translating a **consultant-focused narrative** into a functional dashboard.
- Highlighting **high-impact** apps visually.
- Organizing visuals into **purposeful pages** aligned with a consulting scenario.
- Preparing a PowerPoint to **communicate methodology**.